Abstract

Abstract: R.E.D. Model: Insider Threat Detection AI System

The R.E.D. (Recognize. Execute with purpose. Defend.) Model is a Python-based AI system designed to detect and mitigate insider threats within organizations. By combining rule-based scoring across five risk categories—Information Disclosure, Unauthorized Data Operations, Credential Abuse, Security Evasion, and Software Manipulation—with an optional logistic regression model, the R.E.D. Model generates comprehensive risk profiles for employees. It processes behavioral data, contextual logs, and risk indicators, ensuring data integrity through robust validation of mandatory fields like Employee_ID. The model produces a color-coded XLSX report with detailed risk scores, mitigation recommendations, and a risk score distribution visualization. Key features include scalable chunked processing, customizable configurations, and alignment with the NIST Cybersecurity Framework and 8 industry best-practice strategies. The R.E.D. Model supports real-world use cases such as detecting disgruntled employees leaking data and flagging off-hours vendor activities, providing actionable insights for security teams while supporting compliance and operational efficiency.